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Potential Geomechanics Risks of the Proposed Field Development Strategy in a Faulted and Partially Fractured Stacked Carbonate Reservoir

J. Al Seiri, A. Vantala, H. Al Hashmi, A. Salahuddin, K. Khan, E. Warsito, M. Al-Ghaferi, and M. Al Ajami, ADNOC, Abu Dhabi, UAE; J.J. Guerra and M. Povstyanova, SLB, Abu Dhabi, UAE; M. Haas and A. Hussein, SLB, Geomechanics Centre of Excellence, Crawley, United Kingdom

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Abstract

A 4D geomechanics study was conducted in a heavily faulted and partially fractured stacked carbonate reservoir with the purpose of assessing the integrity risks associated to the field development campaign, characterized by different strategies of hydrocarbon production, and water and gas injection (WAG). The study focuses on the assessment of risks associated to intra-reservoir communication, caprock and reservoir integrity, reservoir deformation and compaction, surface subsidence and uplift, fracture and fault deformation and reactivation, and wellbore stability analyses. The integrity analysis was possible by building a calibrated 4D fluid-flow coupled geomechanical model, that uses the available dynamic models for each reservoir, from which changes in pressure were extracted along the history and forecast. The coupled simulation included stress-strain computations, deformation and failure analysis to assess the potential geomechanics risks of the proposed field development plan (FDP). The coupled geomechanics simulation showed no pore collapse in the different reservoirs but minor elastic deformation barely affecting the porosity and permeability of the intact rock, because of the high rock strength and relatively small pressure changes experienced during the history and FDP. Nonetheless, the fracture network shows a significant deformation, impacting aperture, hence fracture porosity and permeability. This phenomenon might eventually impact the flow patterns and repositioning of infill wells for both production and injection. Furthermore, some fault reactivation potential was observed in specific regions, characterized by heavy faulting, potential presence of natural fractures and subjected to drastic pressure changes. Reservoir deformation same as surface subsidence/uplift was minor, while drilling integrity might be affected by the reactivation of nearby critically stressed fracture networks and faults. Well and completions integrity seems not be at risk due to the high rock and cement strength, and casing design, limiting the amount of plastic deformation and potential failure to near zero, for the conditions of the proposed field development strategy.

Introduction

The field studied is heavily faulted and partially fractured carbonate reservoir divided into three main blocks, with the presence of natural fractures but it is considered a single porosity reservoir. The field is formed

by seven independent reservoirs containing oil, gas, water and CO₂ at different levels. These reservoirs might experience communication at some point during the field development; for this reason, it is necessary to evaluate the potential effects of the production and injection strategies into the single stacked reservoir model (SRM), via geomechanics analysis of rock's deformation and failure.

The carbonate sequence is located at depths of 8000-9000 ft deep with varying thicknesses in the range of 40-60 ft. Due to the textural nature of the rock (high porosity and fractures) it is expected some reservoir deformation during the reservoir development, leading potentially to rock failure and changes in the petrophysical properties, productivity and eventually intra-reservoir communication, especially if the reactivation of faults and/or pre-existing fractures networks occurs.

The database of this study includes 30 offset wells, with ten of them having advanced logs, like sonic and processed wellbore images. There are also rock mechanics tests, pressure and stress measurements in several wells used to calibrate the geomechanics model from pre-production to present day (Noufal, 2020). The 3D dataset includes seismic inversion outputs and dynamic models for each of the 5 reservoirs conforming the SRM. Furthermore, observations and production results of some wells helped to support some of the findings related to the absence of reservoir deformation but potential intra-reservoir communication through the fault frame and fracture network.

Geomechanical Model Construction

The study involved the construction of the geomechanics grid, after embedding the reservoir grid. Then a property modelling workflow was implemented, based on 1D mechanical properties, 3D seismic attributes and static properties. The next step was the pore pressure mapping from the different dynamic models, for 14 selected time/stress steps along the history and forecast. Seismic attributes, static properties, fault pillars and DFN patches (simulated using a Natural Fracture Prediction algorithm, NFP) were also mapped into the geomechanics grid. Intact rock, fractures and fault properties were assigned to the different materials. Mapped pressures and boundary conditions were used to simulate and resolve the 3D stresses and deformation for every coupling step. The resultant equilibrated stresses and strains, same as the predicted failure and displacements were then used to analyze the integrity of the reservoir, caprock, fault, field, ground level, wellbore and completions.

3D Mechanical Earth Model (pre-production)

A 3D Mechanical Earth Model (3D-MEM) was built in the area of study, using structural, static, and dynamic models of the field, along with 1D mechanical earth models (1D-MEMs) tied to lab data. The different static models compromising the reservoir units were merged into a single Stacked Reservoir Model (SRM) to assess the codependency and interaction between the reservoirs during the Field Development Plan (FDP). The 65.6 million cells geomechanics grid was constructed after embedding the SRM with the overburden, sideburden, and underburden zones, usually added to transmit the loads to the reservoir, inducing vertical and horizontal deformation in all the formations (Figure 1).

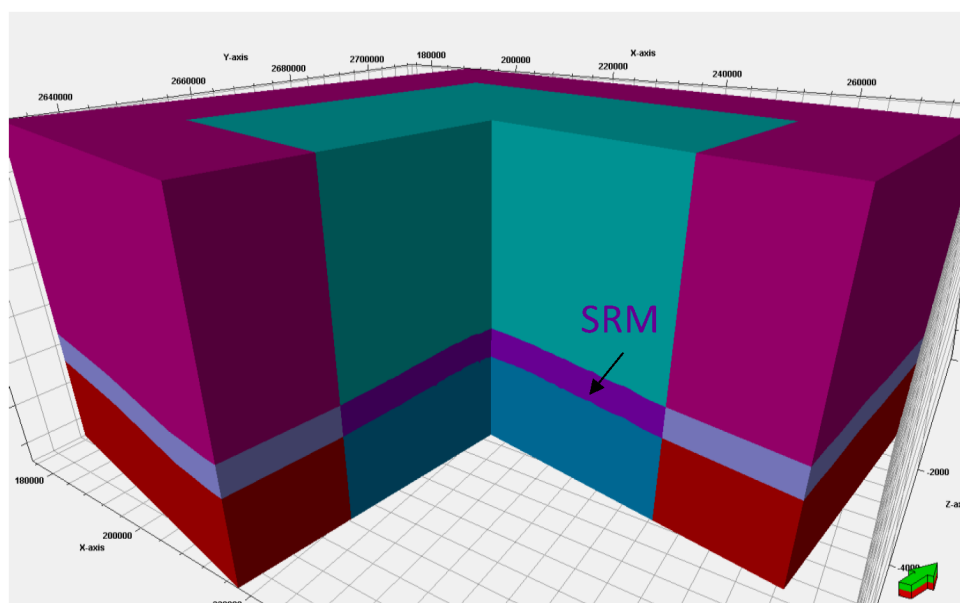


Figure 1—Geomechanics grid after embedding the SRM (purple) into the overburden (green), underburden (dark blue) and sideburden cells (pink, light blue and red)

Mechanical properties (elastic and strength) were computed from calibrated 1D Mechanical Earth Models (1D-MEMs) propagated into the geomechanics grid via petrophysical modelling guided by seismic attributes and static properties as secondary variables. Faults and Fractures were Mapped into the grid with mechanical properties assigned per cell. They were modeled using equivalent material formulation and Mohr-Coulomb failure criteria to predict failure. Finally, pore pressures obtained from dynamic simulation of each reservoir were mapped into the geomechanics grid and then merged across 15 time/stress steps (from 1990 to 2055).

The last component of the 3D-MEM construction is the initial stress state (principal stress magnitudes and directions) which was numerically computed using a finite element simulator with the aim to equilibrate stress and strain in the 3D space, after accounting for the different gravity, tectonic and pore pressure loads, and the different materials (rock and discontinuities) with varying mechanical properties across the field. One of the main components of this initial stress state is the maximum horizontal stress direction (SHmax azimuth) which in the area of study corresponds to N60°W (based on local observations from wellbore images), almost perpendicular to the regional N40°E, dominant in almost all Abu Dhabi area (Noufal, 2016). Finally, the simulated 3D-MEM stresses were calibrated and validated against the 1D-MEM stresses, while the strength and failure models were validated via back analysis (mud window, breakouts and induced fractures analysis). In general, the 3D pre-production stress field showed good alignment with observed wellbore data, and it is ready for one way coupling, for each of the coupling steps along the FDP.

4D-MEM: One-Way Coupled Geomechanical Simulation (history and forecast)

A one way coupled geomechanics simulation was set up to simulate the stress/strain evolution along the history and forecast, additional to the pre-production step (Settari, 1998). The coupled simulation constitutes the 4D Mechanical Earth Model (4D-MEM), used to understand the dynamic behavior of the stress and strain in the subsurface, and its impact in rock deformation and eventual failure (Koutsabeloulis, 1998). Nine time/stress steps in the history were used to calibrate the 4D-MEM and five time/stress steps in the forecast were used for the field integrity risk predictions. Similarly to the pre-production phase, a stress calibration is also conducted for the history, with 4D stresses honoring the 1D stresses and measurements.

A failure analysis was also conducted to validate the drilling observations for all the wells in the history, confirming the validity of the 3D model over time and ensuring the forecasting capabilities.

The 4D-MEM showed stress changes associated with depletion of two reservoirs and injection occurring in other two. One of the reservoirs had both production and injection, mostly water alternating gas (WAG). The last two reservoirs didn't exhibit any pressure change in history but in the forecast, as they will be subject to future development (Figure 2). As consequence of the pressure changes, stresses were also affected, with a minimum horizontal stress (S_{hmin}) mirroring the pore pressure trends, following the poroelastic theory (Terzaghi, 1948, Biot 1955), this is, a drop in the minimum stress during production and an increase during the injection phase.

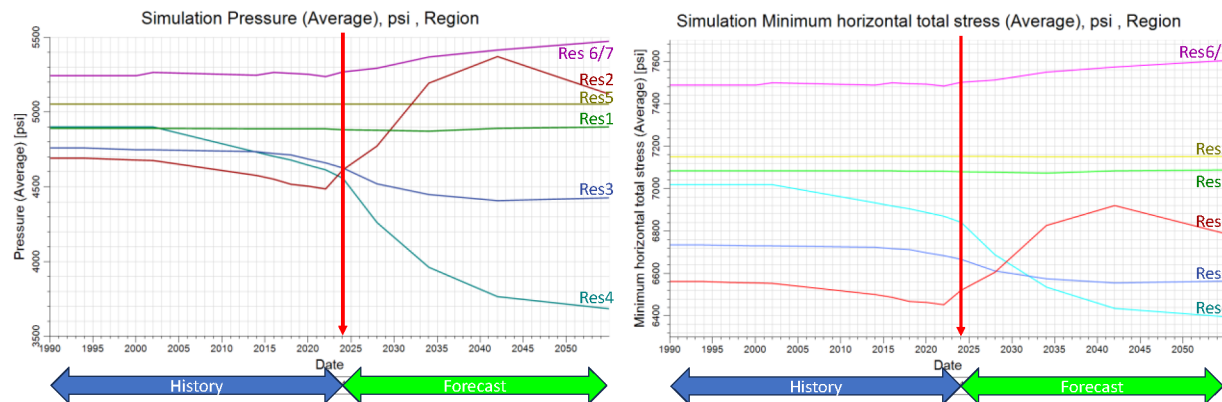


Figure 2—Average pore pressure (left) and minimum horizontal stress (right) for each reservoir along the history and forecast, including production and WAG injection cycles

Potential geomechanics risks of the Field Development Strategy

The 4D MEM is essential to understand and predict the potential geomechanics risks of the proposed Field Development Plan (FDP). Aspects like reservoir, caprock, fault, field, borehole, completions and ground level integrity can be assessed at different times in the forecast, after analysing the 4D stresses, the pore pressure (PP) variations and the resultant deformations (strains) in combination with the appropriated failure criteria to predict rock deformation and failure.

The varying stresses and pressures lead to reservoir deformation and potentially to intact rock failure, and slippage/reactivation of natural fractures and faults. In fact, the amount of rock deformation (both elastically and plastically) might lead to compaction or expansion of the pore structure, and the increase or decrease of the fault/fracture aperture, including in some cases dilation (Vajdova, 2004), which might potentially affect the fluid flow during production and injection operations (Martin, 2018), and even the risk of losses during drilling. All these effects might occur in the caprock, reservoir, faults, fracture network, wellbore, completions (cement and casing) and even in the ground level, affecting the integrity of some of these regions.

The following geomechanics attributes are critical to assess these potential integrity risks:

1. Stresses and Pressures (minimum stress S_3 , pore pressure PP, bottom hole pressure BHP), used to determine the risk of tensile failure and hydrofracturing.
2. Yield Mode (YELDMOD, FLT_MODE, DFN_MODE), used to determine if the rock is failing in tension, shear, or pore collapse.
3. Yield Values (FVAL, YIELDVAL, YLD_xx_x, FLT_YLDV), used to determine the distance to failure or yielding for fractures and faults.

4. The different forms of strain (VOLUMETRIC, STRAIN_{xx}, PLSTRN_{xx}, FLT_E/P/N/S strain, DFN_E/P/N/S strain), used to determine the amount of elastic and plastic deformation (severity of the deformation and yielding).
5. Displacements (DISPL_ZZ), used to determine the amount of rock displacement in the vertical direction (Z) in both reservoir and ground level.

These attributes can be used single or in combination to assess existent or potential integrity risks during the FDP, including the following applications:

- Reservoir Deformation, affecting the (Petrophysical changes and displacement)
- Ground level displacement (upward/downward movement)
- Reservoir and Caprock Integrity (Failure & Fracturing)
- Fracture Deformation (Petrophysical changes and displacement)
- Fracture Stability (Reactivation potential)
- Fault Integrity (Reactivation potential)
- Drilling Integrity (Wellbore Stability)
- Well Integrity (casing, cement and open hole completions)
- Pore Collapse (cap failure)

Next sections detail the results of each of these integrity risks for the area of study (excepting only the Well Integrity).

Reservoir and Caprock Integrity risk

There are two key attributes used to assess reservoir and caprock integrity. One option is the YIELDMOD property, which showcases the integrity analyses and rock/formation failure. It states the type (mode) of failure and is a discrete attribute, while the YIELDVAL property is a quantitative property as a stress attribute. It has the units of stress and is defined such that "zero" and positive values imply failure, whereas negative values refer to no failure (stable). This distance to failure attribute is measured perpendicular to the Mohr Circle and the failure line is approximated as a linear curve. The YIELDMOD attribute is presented in Figure 3 for the reservoir rock.

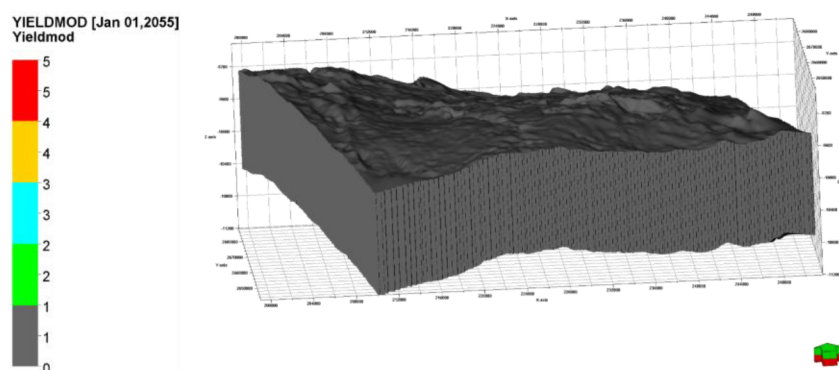


Figure 3—YIELDMOD for the entire 3D geomechanical model, showing no rock failure (0, grey).

The second option is to present integrity analyses by using the Mohr-Coulomb failure diagrams and P-Q plots for different representative elements or grid cells in key zones or K-layers. The Mohr-Coulomb

depicts how far the stress state deviates to the point of failure, using the Mohr-Coulomb failure criteria, i.e., accounting only for the maximum and minimum principal stress (Labuz, 2012). The P-Q diagram shows the stress path along which the combined effect of pressure and temperature on the stresses can be interpreted and is hence a representation of the Mohr-Coulomb stresses. Typically, the Mohr-Coulomb analysis is conducted on representative grid cells near the injection or production zones, where the largest variations in pressure are expected. It is also used along the faulted or fractured elements to determine the stability of the fault and fracture networks.

Figure 4 shows the Mohr-Coulomb and P-Q plot for a representative cell in the reservoir section. Note that for the intact rock case, there is no chance of failure along the history and forecast. In general, no tensile or shear failure observed in the intact rock for the reservoir or caprock zones. Deformation is strictly elastic, with some volumetric changes in the reservoir rock. Mohr-Coulomb and P-Q plots confirmed stability across all time steps in the forecast (2024 through 2055), with the Mohr circles and stress path in blue far apart from the failure line in red. Additionally, all the deformation in the model was elastic, with negligible plastic strain across the grid (<0.00006), indicating high rock strength and moderate stress changes are occurring during the forecast.

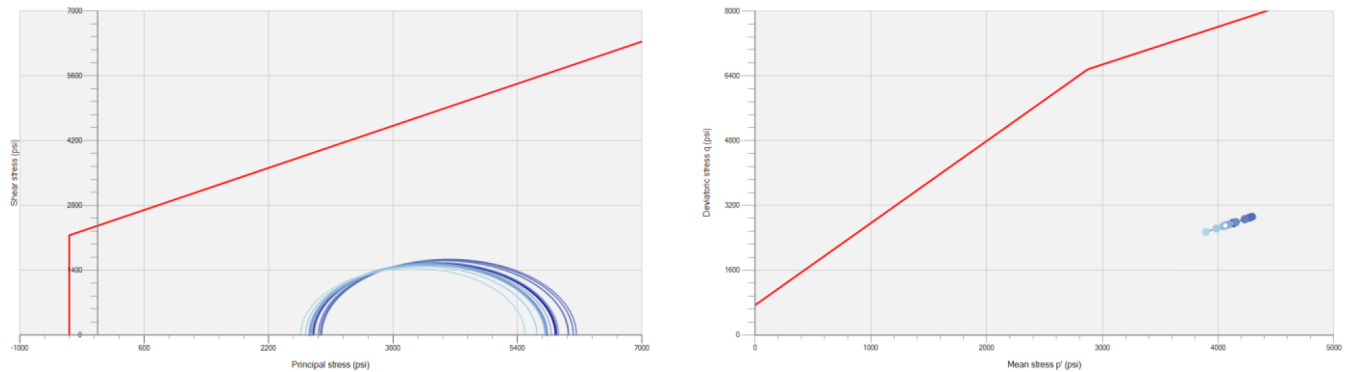
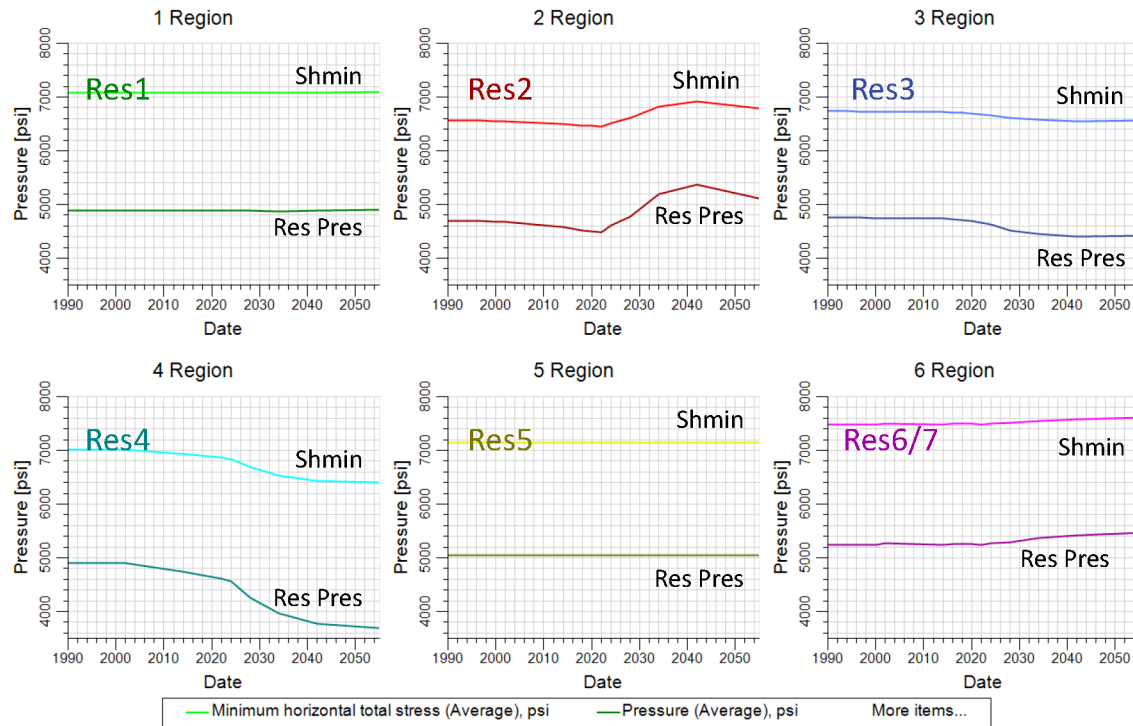


Figure 4—Mohr circles (left) and P-Q stress plot (right) for a representative grid cell, showcasing no intact rock failure for all time/stress steps

The risk of tensile failure is addressed in Figure 5. Note the high minimum horizontal stress relative to the average reservoir pressure, suggesting that the risk of hydrofracturing is low, even during injection operations. The relative distance (window) between the minimum horizontal stress and pore pressure ($S_{hmin}-PP$) is large in all the reservoirs, ranging from 1,500-2,700 psi. The narrowest window occurs in reservoir 2 and the largest window exist in reservoir 4, ranging from 2,100-2,700 psi.



Shmin → Minimum Horizontal Stress P → Reservoir Pressure < Average >

Figure 5—Minimum horizontal Stress and Pressure evolution per reservoir (Average). Shmin-PP injection window

Reservoir Deformation risk (matrix porosity and permeability changes)

Reservoir deformation can be assessed by analyzing the changes in volumetric strain. This parameter englobes changes in the six (6) components of the strain (XX, XY, XZ, YY, YZ, ZZ), used to compute the volumetric strain change, ε_v , representing the dimensionless volumetric deformation of the rock volume (the bulk cell volume) from one step to another. The ε_v is calculated as follows:

$$\varepsilon_v = \varepsilon_{xx} + \varepsilon_{yy} + \varepsilon_{zz}$$

Where: ε_{xx} --> Strain in XX, ε_{yy} --> Strain in YY, ε_{zz} --> Strain in ZZ

The volumetric strain changes are primarily attributed to the pore space given that the mineral compressibility is significantly much smaller compared to the rock skeleton compressibility (the pore structure), typically in between 1 and 2 orders of magnitude in difference. The volumetric changes in the reservoir units will produce changes in the matrix porosity and permeability than can be translated into porosity and permeability multipliers that will impact the original petrophysical properties, making them dynamic rather than static during the life of the field.

Positive and negative changes of matrix porosity and permeability (compaction and expansion) are observed in the different reservoirs at present and future date in 2024 and 2055 respectively (top Figure 6), while mostly positive changes (compaction) occur in some of them. These results align with the production and injection campaigns (Xi, 2018). In general, the matrix porosity multiplier varies between -1.0 to +0.5%, most of it happening at the end of the FDP. In a similar way matrix permeability multiplier varies between -1.8% to +1.3% at the end of the FDP, slightly higher than the porosity multipliers, as expected (bottom Figure 6).

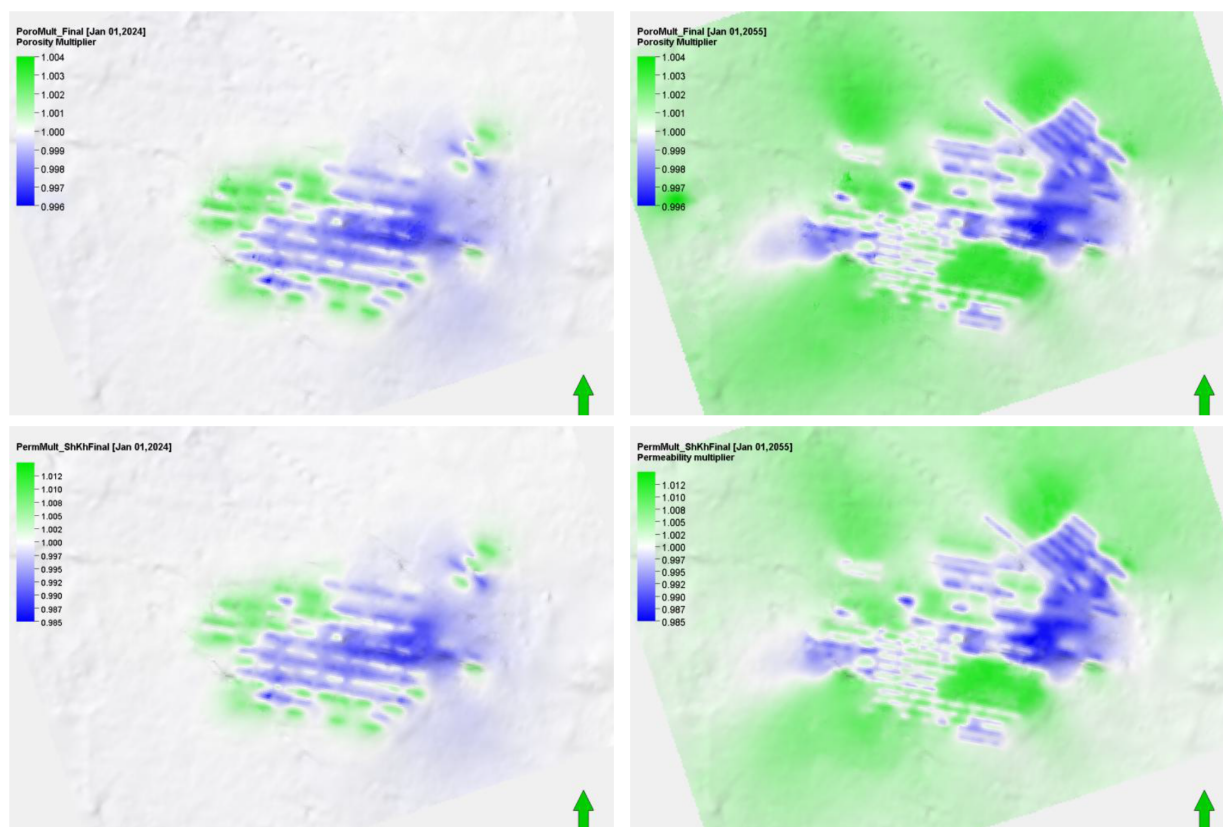


Figure 6—Matrix porosity changes (above) and permeability changes (below) displayed as porosity and permeability multipliers for reservoir-1. Green zones highlight an increase while blue areas indicate a decrease in the multipliers.

The compaction analysis for poro-perm changes let us conclude that relative porosity and permeability changes across the matrix are negligible. The pressure changes do not deform the rock matrix in a significant way, limiting the stress-dependent variation of the productivity and injectivity of the wells; nonetheless, the stress-dependent contribution of the fracture system might play a significant role in the total permeability change, possibly impacting the fluid flow and hydrocarbon production.

Pore Collapse risk

Pore Collapse is a catastrophic failure that reduces drastically the rock porosity and negatively affects permeability after the pore structure is destroyed and the grains disaggregated (Dudley, 2005). This phenomenon can occur in highly porous and weak sands, a few poorly cemented sandstones, chalk and some grained limestones (calcarenite, coquina, grainstones, etc). Pore collapse might be prevented by selecting the perforating intervals, limiting the pressure drawdown and managing the reservoir pressure (reducing the depletion rate, doing pressure maintenance, etc).

A geomechanics simulation was launched using a "critical state" failure criterion to assess the risk of pore collapse or cap failure (Fossum, 2000). P0 (collapse pressure) and radius of ellipse (influential radius) were included as additional variables to the Mohr-Coulomb/ Modified Lade criteria. Based on the stress outputs, the plasticity behaviour and potential pore collapse was investigated.

The critical state simulation results produce two important failure results in the form of Yield Values: YLD_FC (distance to cap failure) and YLD_FS (distance to shear failure). Both attributes are largely in the negative spectrum (Figure 7), which implies no cap failure (i.e. no pore collapse) nor plastic deformation is happening in the 3D model, from history to forecast, excepting a small region depicted in Figure 7 for year 2024. There are localised grid cells failing by cap in this region, but they are mostly related to pinch-

outs. The importance of these results resides on the fact that all the potential failed elements are localized far from the planned wells.

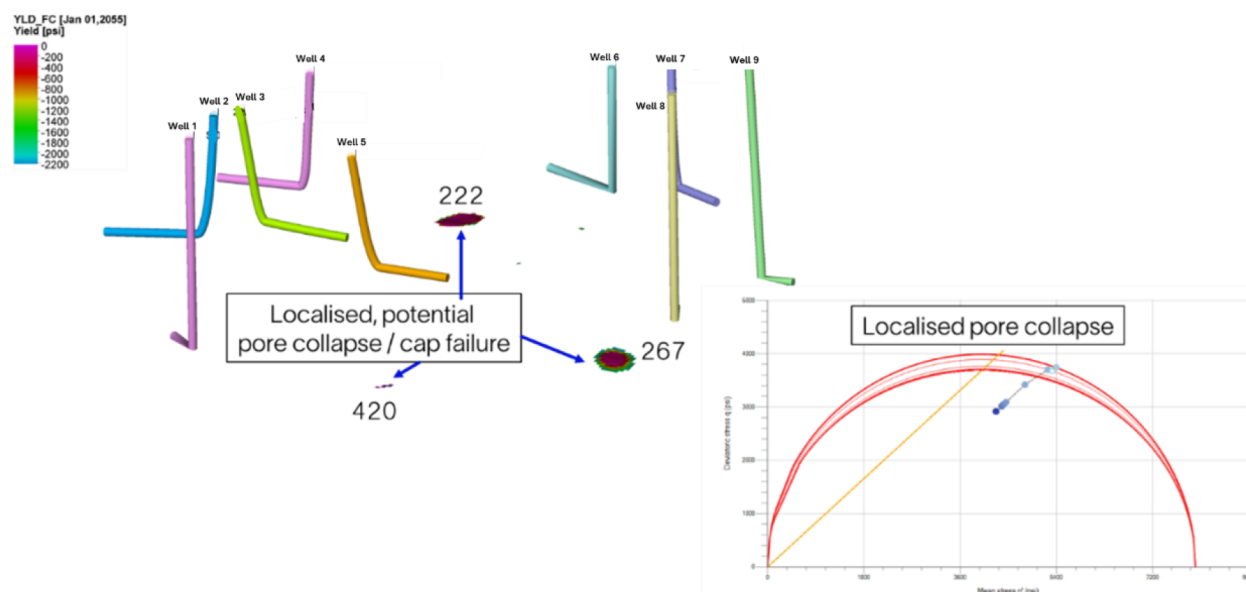


Figure 7—Yield value shown for time step 2055 and planned well, showing potential pore collapse / cap failure in localized grid cells in K-layers 222, 267 and 420. The Critical State Mohr Coulomb failure criteria is shown to the right

In general, the 4D geomechanics model did not show any true pore collapse throughout the field based on the 1-way coupled stress simulations. This result is expected due to the relatively medium to high rock strength (typically 5,000 to 22,000 psi) and pressure changes which are not too drastic (ranging from -2000 to +2000 psi). Indeed, the effective stresses are relatively low (ranging between 3,000 and 6,000 psi), due to the low depletion and continuous pressure maintenance in most of the reservoirs. The porosities are also typically lower than 20% (ranging from 1 and 25 %) so the reservoirs might be classified as low to medium porosity. Usually, cap failure is expected at much larger porosities in weak carbonate rocks, typically larger than 27%, and effective stresses in the high end, typically larger than 7500 psi. None of these conditions exist in the area of study and they are not expected to occur neither.

Fracture and Fault Stability risk

The inclusion of discontinuities like natural fractures in the form of Discrete Fracture Networks (DFN) allows the incorporation of different material affecting the subsurface stress equilibrium in naturally fractured environments. It also enables the analysis of potential critically stressed fractures. The DFN of this study was created using a Natural Fracture Prediction workflow (NFP). It was included implicitly as an anisotropic material affecting the stress and strain distribution of each intersected grid cell. The properties of this anisotropic material were assigned depending on the fracture characteristics, stiffness and density, and its failure will depend on the strength properties, and the shear and normal effective stresses acting on each fracture patch.

Fracture stability and failure can be predicted at 3D level by analysing attributes like Fracture Yield Mode (DFN_YLDMOD), or Fracture Yield Value (DFN_YLDV), or by determining the presence of Critically Stressed Fractures (CSF) from attributes like distance to failure or Coulomb Failure Function (CFF). The concept of fracture and fault stability is depicted in Figure 8, where an arbitrary fracture patch or fault plane (left) is subject to normal and shear stresses (centre) that might exceed or not the failure line in a typical Mohr-Coulomb diagram (right).

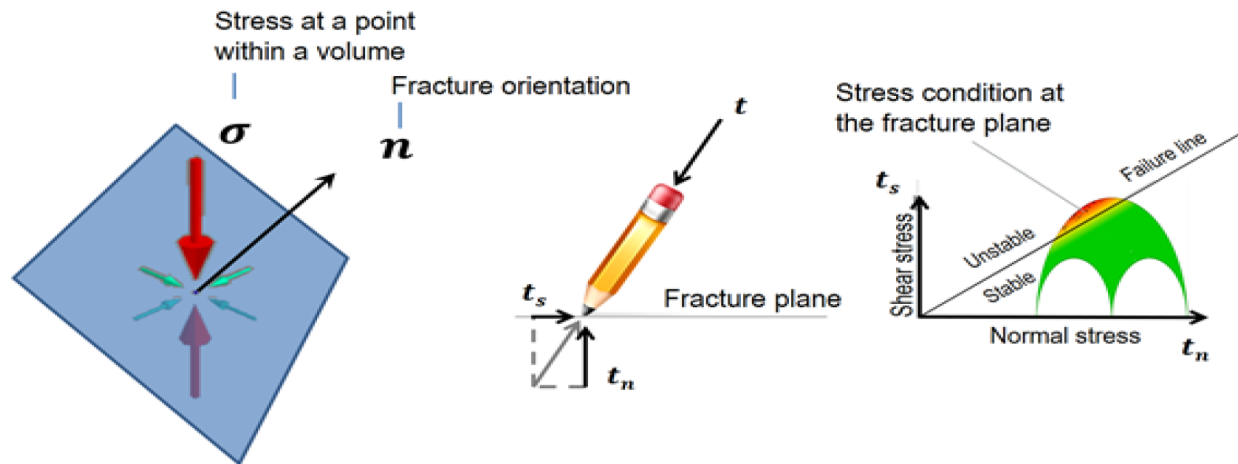


Figure 8—Concept of fracture stability analysis based on Mohr-Coulomb (normal/shear stress) analyses.

Assessing the fracture stability is only one of the key aspects when assessing the natural fracture behaviour during the FDP. A more advanced analysis includes the simulation of the fracture deformation, by analysing the normal and shear elastic and plastic strains (DFN_E/P/N/S strain) that will impact the fracture aperture, porosity and permeability, as will be described later.

Different fracture families (DFN) derived from the NFP prediction were mapped into the geomechanics grid and accounted for in the geomechanics simulation; each one representing a particular fracture family, with its own characteristic and tectonic origin (e.g. faulting, folding, bending), as shown in Figure 9.

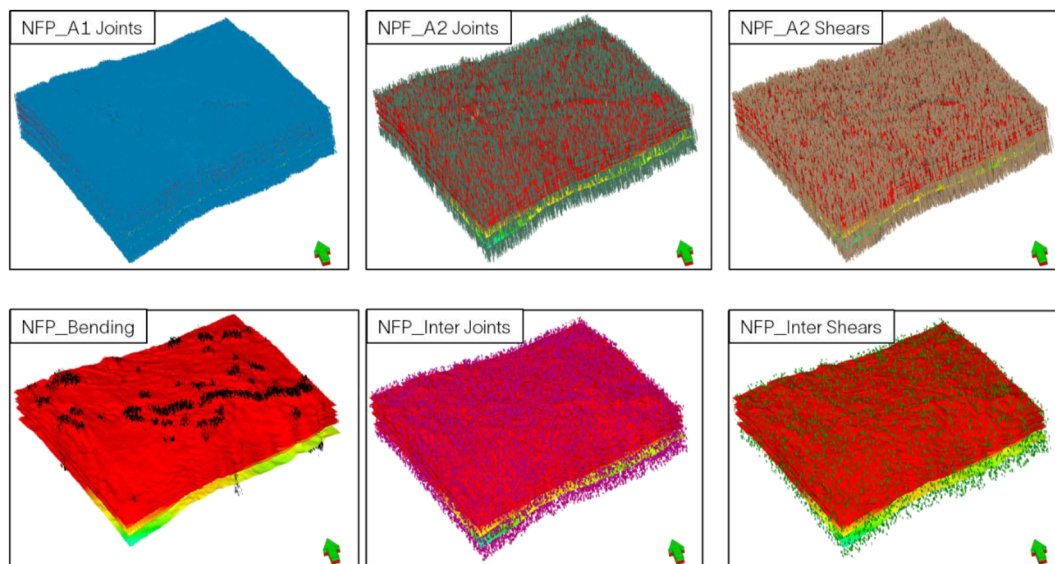


Figure 9—Fracture patches implemented into the 3D MEM simulation from different tectonic origins (only six out of nine predicted fracture families are displayed for illustration purposes).

Presence of Critically stressed fractures (CSF)

The DFN yield value at convergence is used to check whether a natural fracture is critically stressed (i.e. potentially reactivated). The DFN_YLDV values in the field range between minus -3000 to 0 psi; whereas a high negative value depicts points below the failure envelope, which are not critically stressed. Values close to the failure line, like those with a low negative number or even 0 (i.e. exactly on the failure envelope), correspond to the stress conditions of potential critically stressed fractures. Contrary, positive DFN_YLDV

values means the fractures exceeded the elastic limit and deformed plastically (likely dilating and increasing volume). The same concepts apply for the evaluation of fault deformation and reactivation.

The result of the simulated DFN YIELD VALUE in year 2024 are shown in Figure 10, with very few and sparse grid cells close to or at zero, implying the presence of few critically stressed fractures in those elements.

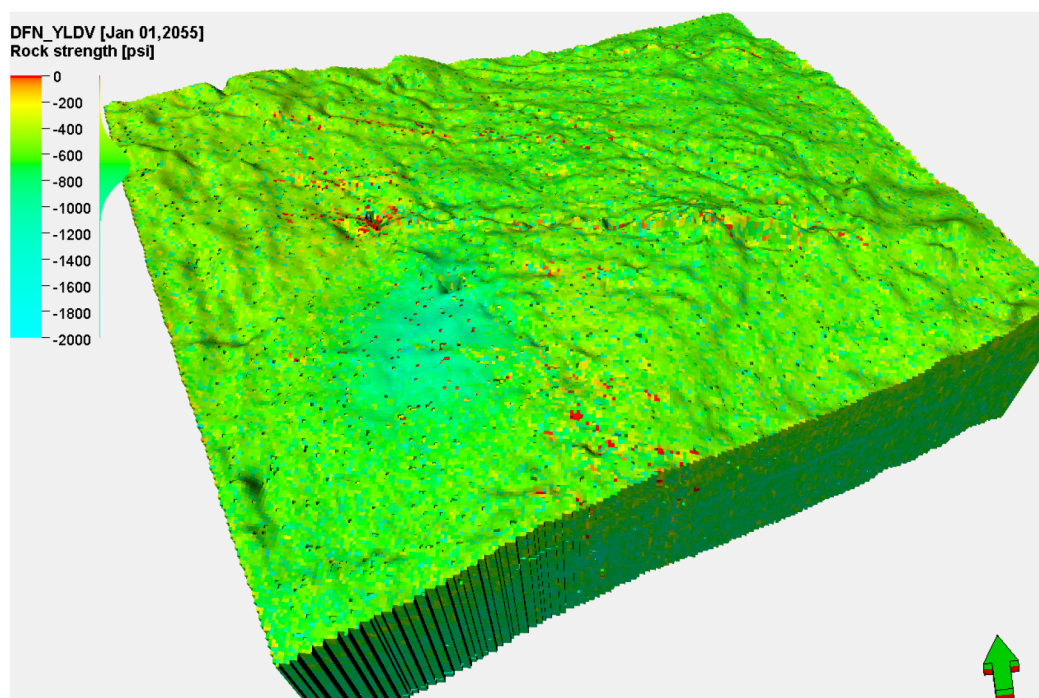


Figure 10—DFN YIELD VALUE assessment for time step 2055 and friction angle of 20 deg.

A 3D view of the spatial location of the potentially critically stressed fractures is shown in Figure 11. Natural fractures in this region show a higher potential of being reactivated (i.e. they are close or above the Mohr-Coulomb Failure Line). Note that most of the critically stressed fracture grid cells occur near the faulted area, which is somehow expected and reflects the consistency of the simulation results. Note that most of the critically stressed fracture grid cells occur around the planned well in green in the center, and some more around the light blue and orange wells in the north and south respectively. These wells might produce an increased productivity/injectivity per foot, if drilling across a fracture network. They would help to validate also the case for the presence of critically stressed fractures in the forecasted regions, supporting the results of the simulation.

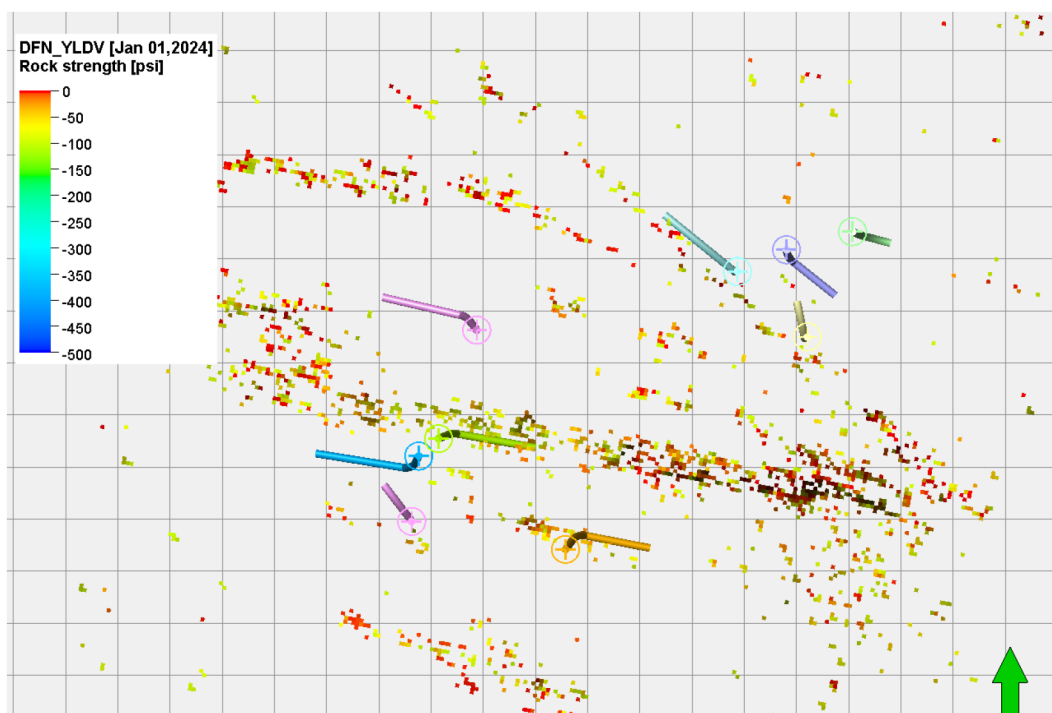


Figure 11—Potentially critically stressed fractures map at time step 2024 (DFN YIELD VALUE > -100 psi).

The presence of critically stressed fractures indicates failure and occurrence of plastic deformation in most of the cases; nonetheless, their absence does not mean anything is happening in the fracture network. In fact, there is deformation occurring in the elastic domain, inducing fracture closing, aperture and even dilation, this phenomenon impacts both fracture porosity and permeability, as will be analysed next.

Fracture Porosity and Permeability changes

The fracture deformation and analysis of potentially critically stressed fractures posed the question of their impact on permeability and fluid flow. Previously, it was shown that matrix porosity and permeability proved to be minor in both history and forecast; nonetheless, the geomechanics simulation revealed that fractures have a significant impact on the field-scale permeability and they might contribute largely to increased or decreased fluid flow. Fracture porosity and permeability multipliers similar to those used for the matrix, were calculated based on the total normal displacement (or strain) and the original fracture aperture derived from interpreted image logs.

The fracture porosity multiplier is essentially the same fracture aperture multiplier, so that the fracture porosity multiplier can be used to obtain the fracture permeability multiplier, using for example a cubic law. The set of equations below (Eq 8 to Eq 11) describe the general concept behind the porosity and permeability multiplier computation based on normal strain/displacement only. Keep in mind this is a quick look approximation to probe the concept, while a rigorous permeability multiplier must be obtained by analyzing both normal and shear components of each fracture patch, producing a fracture permeability tensor, not a scalar, as presented below.

$$\Delta \phi_{fracture} = \frac{\Delta Frac \delta_n}{a_0} * 100$$

$$FracPoroMult = \frac{Frac \delta_n - \Delta Frac \delta_n}{Frac \delta_n}$$

$$FracPermMult = \left[\frac{Frac \delta_n - \Delta Frac \delta_n}{Frac \delta_n} \right]^2$$

Where, $\Delta\phi_{fracture}$: delta porosity fracture (dimensionless); $\Delta Frac \delta_n$: delta fracture normal displacement (m); a_0 : initial aperture (um); $Frac \delta_n$: fracture normal displacement (m); $FracPoroMult$: fracture Porosity Multiplier (dimensionless); $FracPermMult$: fracture permeability multiplier (dimensionless)

The left side of Figure 12 shows the fracture porosity multiplier map in Reservoir-2 at time step 2024. Note the fracture porosity decreases in zones with heavy production (in blue color), and increases typically in zones with some injection (green color). The same analysis conducted in 2055 shows a significant change in the fracture porosity compared to 2024 (Figure 12 to the right). In this case the fracture porosity multiplier oscillates in between 0.8 and 1.3 (equivalent to -20% and +30%) in 2055. These changes will affect fracture permeability, which is function of the aperture and variations in the normal and shear displacement. In general, fracture porosity multipliers for all the reservoirs oscillates in between 0.88 and 1.1 (equivalent to -12% and +10%).

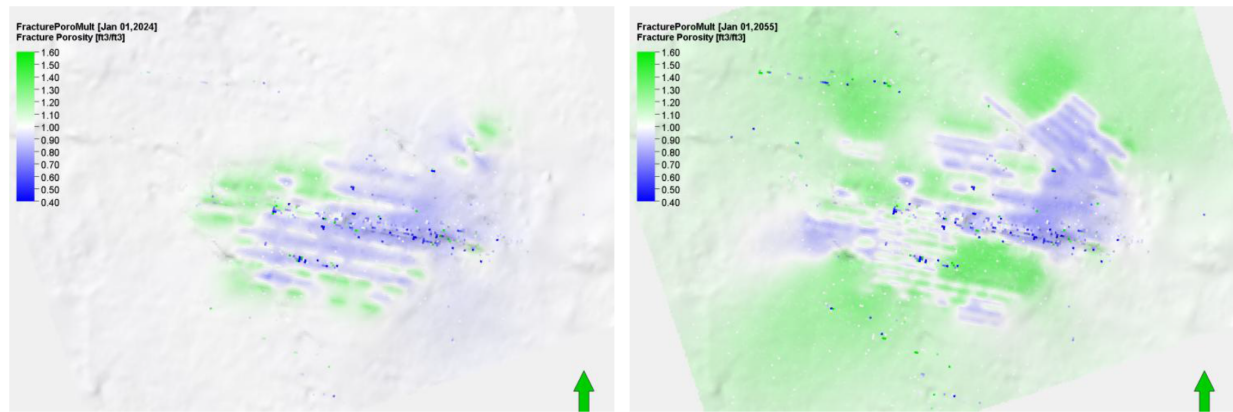


Figure 12—Fracture porosity multiplier for Reservoir-2 in 2024 (left) and 2055 (right). Increase in green, decrease in blue.

Computing these multipliers will not only help in visualizing the impact of the stress changes in the porosity and fracture permeability but more importantly, they will affect the fracture porosity and permeability of each cell in a dual porosity/dual permeability model (rather than the single porosity model that currently exists in the field), affecting the fluid flow, pressures and production response.

Figure 13 show the fracture permeability multiplier map in Reservoir-2 at time step 2024 (left) and 2055 (right). Note the fracture permeability decreases in zones with heavy production (in blue color), and increases typically in zones with some injection (green color), following same pattern of the fracture porosity multiplier. The fracture permeability multiplier for all the reservoirs oscillates in between 0.78 and 1.14 (equivalent to -22% and +14%). The same analysis conducted in 2055 shows a significant change in the fracture permeability multiplier compared to 2024, where the multiplier oscillates in between 0.6 and 1.7 (equivalent to -40% and +70%) in 2055. These changes will surely impact the fluid flow, hydrocarbon production and potential location of infill wells.

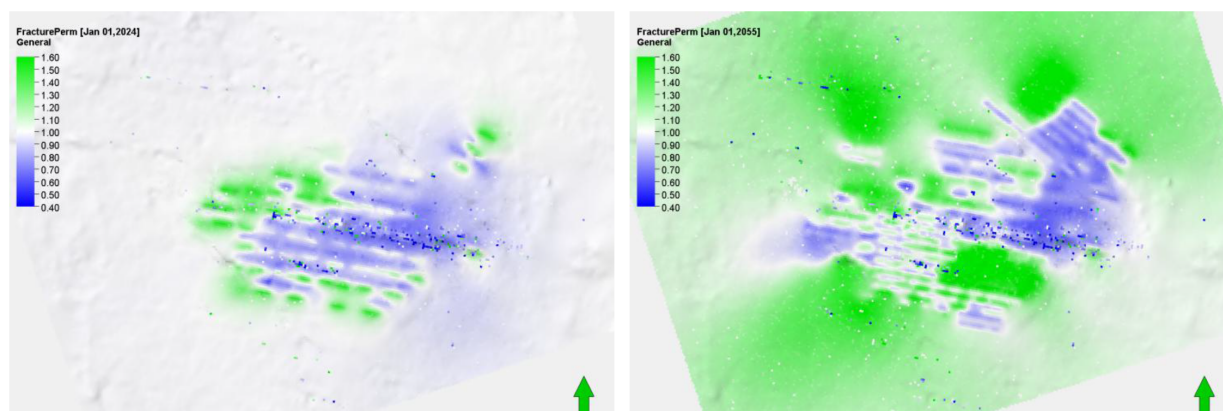


Figure 13—Fracture permeability multiplier for Reservoir-2 in 2024 (left) and 2055 (right). Increase in green, decrease in blue.

The multipliers evolve as function of the depletion and injection operations, decreasing and increasing accordingly. Thus, the fracture permeability mostly decreases when no injection occurs. Conversely, in regions where injection occurs, the porosity and permeability values near the original conditions and even slightly higher. In general, it can be said that fracture permeability is increased by up to +70% in the areas with heavy injection and reduces by up to -40% in areas with heavy depletion. This pattern continues over time, being maximised at the end of forecast in 2055. Furthermore, the regions with increase in fracture permeability follows the location of injector wells, while the decrease is associated to producer wells nearby. This also suggests that some fractures nearby injector and producer wells might be critically stressed and it might contribute to enhanced fluid path for injection or production.

The expected theoretical change in fracture permeability values can reach the order of 100 to 300 Darcy (in sweet spots up to 2,500 Darcy), suggesting it might be necessary to create a dual porosity / dual permeability fluid-flow model in the future, run the two-way coupling including the dual model and assess their effects in the fluid flow patterns, hydrocarbon production, dynamic pressures and stress/strain simulations.

Fault Reactivation potential during the FDP

Faults are treated as weaker zones from a continuum mechanics perspective. To evaluate the fault stability, all effective stress tensor components acting on the faults and reservoir pressure must be extracted for each time step. Characterizing this stress state is crucial to analyze the reactivation potential and mechanical behavior during the field development. Fault reactivation may cause early water-cut and create a risk of potential water/gas/oil leaking from injecting zones. Furthermore, significant localized deformation near faults may cause wellbore deformation and instability if the well trajectory is close to the deformed zone.

A pre-existing discontinuity, such as a fault, is reactivated when the critical shear stress on the fault plane is exceeded under a certain normal stress, provoking the slippage along that plane; this is the same concept of the fracture stability analysis, but this time accounting for fault properties and stresses around those faults, whose magnitudes and directions are affected by the fault properties and filling material (gouge). The simulated in-situ principal stresses are converted to normal and shear stresses acting against the different fault planes crossing the cell elements. If the critical shear stress on a fault is not exceeded, it is referred to as stable under the current stress and pressure state and the likelihood to be reactivated is proportional to the distance to failure, as depicted in Figure 14.

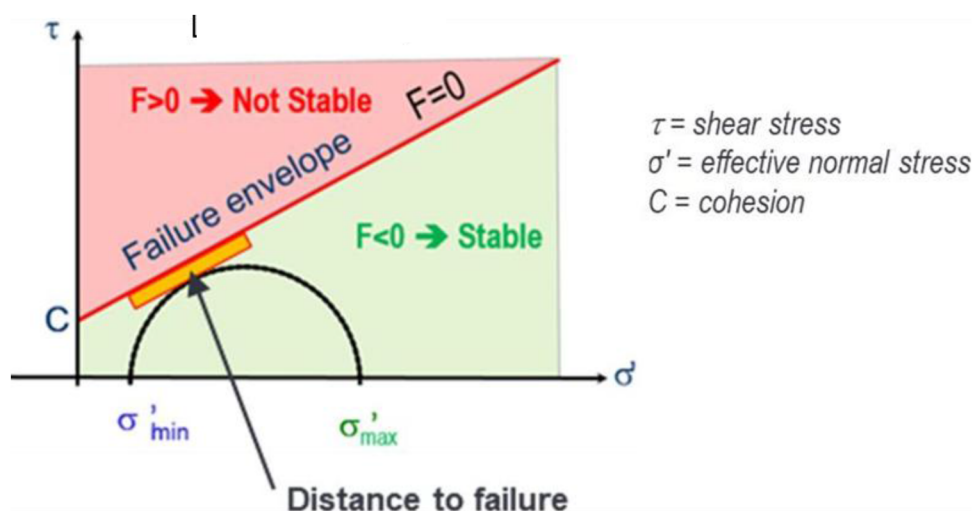


Figure 14—Concept of fault reactivation and "distance to failure" property (F) using a Mohr-Coulomb failure criterion.

The stability of the fault can be assessed by simulating the Fault Yield Values (FVAL or FLT_YLDV, aka distance to failure), similar to that obtained for the intact rock but assessed for the specific orientation of the fault plane relative to the stress tensor orientation, and the fault strength properties. The Fault yield Value has the units of pressure and is defined such that "zero" and positive values imply an unstable condition. Other outputs of the analysis include the Fault Yield Mode (FLT_MODE), the presence of Critically Stressed Faults patches (CSF), Critical Pore Pressure (CPP), Critical Pore Pressure Change (CPPC), Slip Tolerance (ST) and Coulomb Failure Function (CFF). Fault stability can be assessed using the coupled model outputs but uncertainties must be kept in mind (Yoon, 2024).

The fault stability and reactivation potential in the area of study was computed for each time-step, using the available fault framework model (composed of hundreds of high angle faults segments intersecting the different reservoirs), and after including the prevailing fault properties and orientation. The calculation also accounted for pressure changes post-depletion or post-injection, and the stress field tensor and direction to obtain the new shear and normal stresses resolved on each fault segment. In general, areas with active injection based on the dynamic pressure model are likely to be reactivated, especially in normal faulting systems. The same might happen in areas with high depletion and reverse faulting systems. Fault stability was then calculated based on the Mohr-Coulomb criterion after considering the fault properties (friction angle and cohesion). The fault stability analysis helped to determine the stress state on the faults, location of critically stressed faults (CSF) and conditions required for slippage (reactivation).

The maximum horizontal stress direction across the field was estimated as N60W-S60E, with light rotations due to the presence of the faults. In fact, detailed studies, for certain layers and regions reveal small stress redistributions across the field that, nonetheless, differ significantly with the regional stress direction. This local vs regional stress direction uncertainty required a stress direction sensitivity analysis, because this variable not only impact the fault reactivation potential but also the well design. In fact, wells drilled parallel to the minimum horizontal stress (SHmin) might be not only more stable in terms of drilling, but also they might intersect a larger number of natural fractures in the field and produce transverse fractures during a hypothetical hydraulic fracturing campaign.

Figure 15 shows the field wise fault reactivation potential using the Coulomb Failure Function (CFF) filtered from -200 to 0 psi at time step 2024 (the Fault Yield Values - FLT_YLDV attribute can also be used for this purpose), where the location of potential critically stressed faults (CSF) correspond to those elements where CFF or FLT_YLDV approaches to zero (note both CFF and FLT_YLDV are negative for stable faults). All values equal or above zero mark the onset of the reactivation potential. Larger CFF and FLT_YLDV are observed near the injector wells.

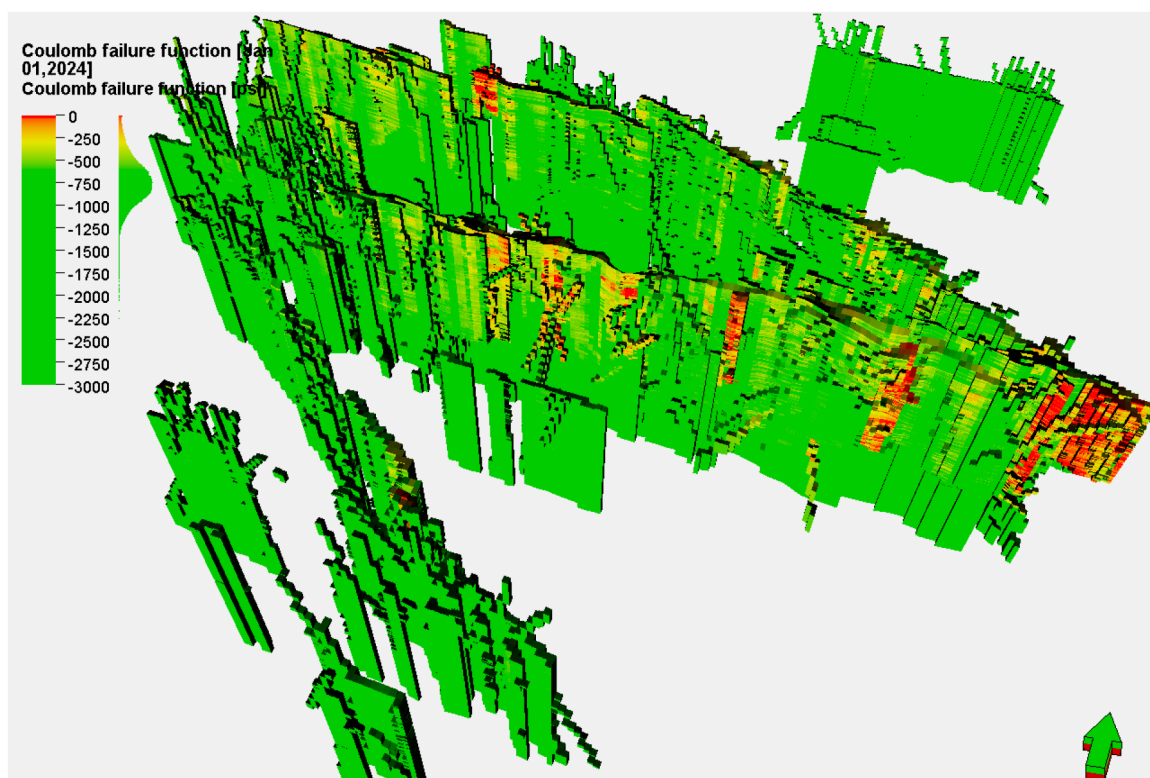


Figure 15—Fault reactivation potential filtered for a CFF $\Rightarrow$ -200 psi. Faulted elements in red and yellow mark a potential for fault reactivation.

The filtered CFF values in Figure 15, highlighting the fault grid cells that are at or close to reactivation. The green cells represent stable conditions, the coloured cells show a higher potential for reactivation with red being the maximum reactivation potential. The CFF analysis shows that there is fault reactivation potential in specific regions in the center of the development area. In those regions, the reactivation potential extends vertically along the reservoirs from Reservoir-2 down to Reservoir-6, creating a continuous connection among some of them, and hence increasing the likelihood to be reactivated. This is typically associated with the increase of the injection and reservoir pore pressure in some regions, as occur in the case for faults close to injector wells.

The presence of CSF might impact the intra-reservoir isolation/communication and even the borehole stability. The first was observed in a couple of producing wells during the history, showing correlation between the presence of CSF and the interference/communication between reservoirs. Unfortunately, none of the offset wells crossed CSF to validate the wellbore instability risk. Despite this, this risk can be warned by looking at CSF regions like the one highlighted in Figure 16, where the planned light green well is too close to a faulted elements that might be critically stressed, potentially leading to intra-reservoir communication and even wellbore stability problems like wellbore wall collapse and mud losses.

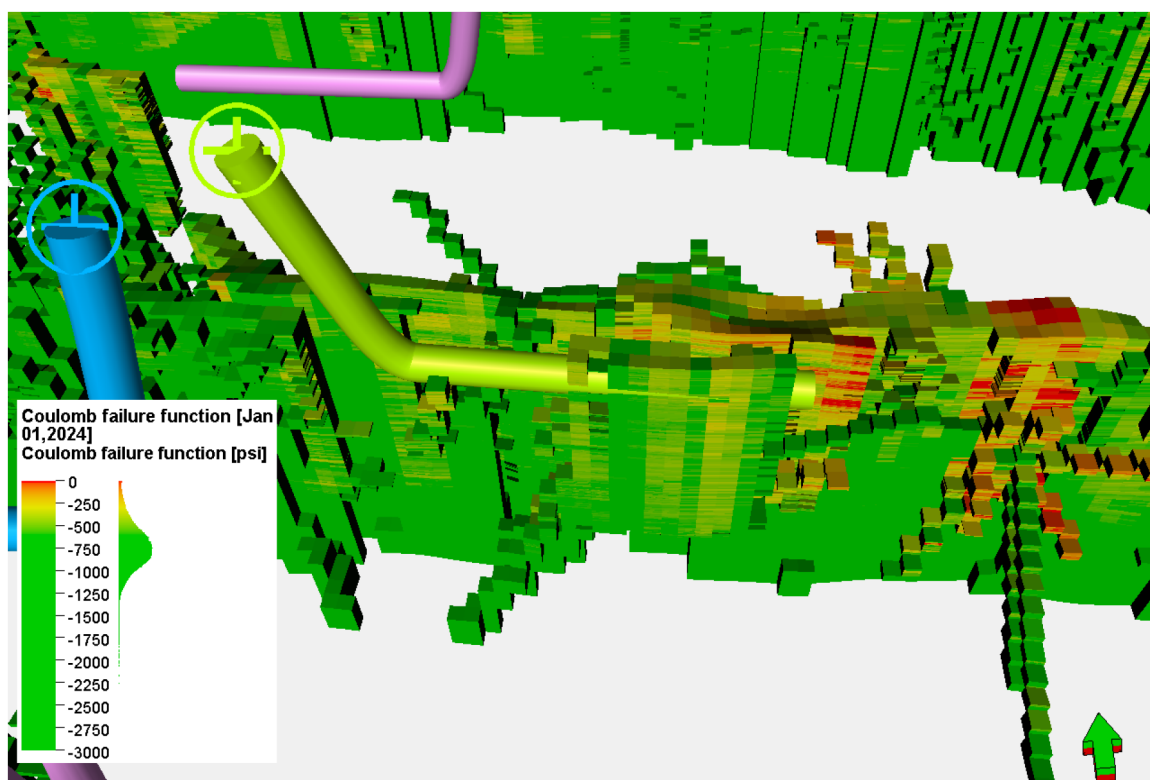


Figure 16—Fault reactivation potential near planned green well, crossing grid cells with large CFF values (in yellow/red).

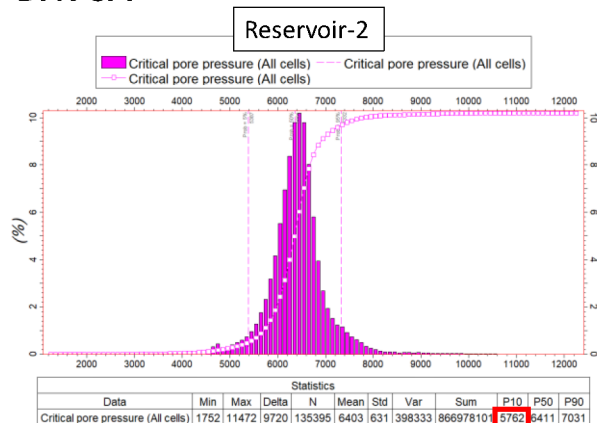
Maximum Injection Pressure Analysis

As mentioned before, the risk of fault reactivation under current depletion/injection conditions is low; nonetheless an increase in reservoir pressure impose risks associated to the creation of tensile fractures, hydrofracturing, shear fractures and/or the reactivation of existing weak planes and discontinuities. In fact, natural fractures and faults constitute the weakest point in terms of reservoir and caprock integrity in a naturally fractured environment, and any local increase in pore pressure might convert a stable discontinuity into a critically stressed one, inducing a potential leaking point or even a preferential path for vertical fluid flow and intra-reservoir migration.

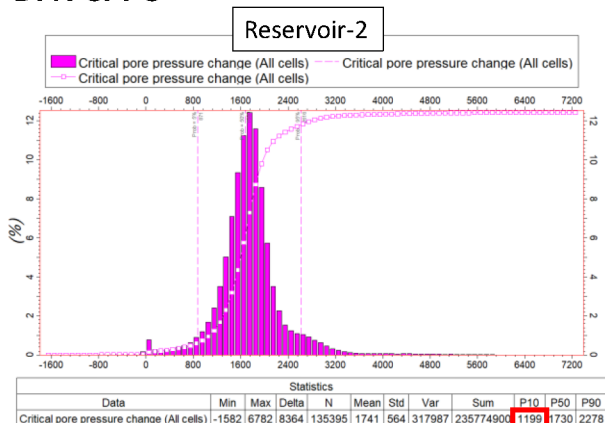
In this sense, the analysis of attributes like Critical Pore Pressure (CPP) and Critical Pore Pressure Change (CPPC) help to determine the maximum injection pressure threshold required to destabilize the fault system on a cell basis, this is when the critical stress is achieved, and fault slippage or reactivation occurs. These attributes combined with the injection pressure window (defined as the difference between the minimum stress and the pore pressure) can be used to determine what occurs first: hydrofracturing or the reactivation of existing discontinuities (faults or fractures).

The Critical Pore Pressure (CPP) and Critical Pore Pressure Change (CPPC) were computed in the area of study for both DFN and Faults, as presented in the histograms of Figure 17 for Reservoir-2. The histograms include the statistic (with percentiles), the cumulative distribution function (CDF) and P5, P50, P95 quartiles derived from the CDF. For purposes of analysis the P10 values of CPP and CPPC were taken as a conservative scenario (although not the worst case) for fractures and fault reactivation potential. These values correspond to the minimum possible pore pressure required to reactivate a significant number of fractured/faulted elements (approximately 10% of the population). In a similar way, the minimum possible pore pressure change was computed at P10 percentile.

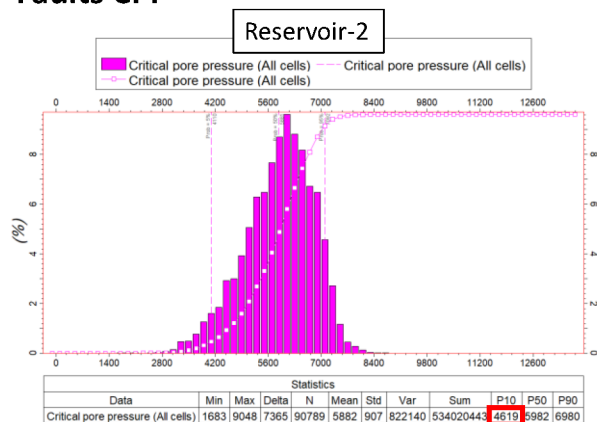
DFN CPP



DFN CPPC



Faults CPP



Faults CPPC

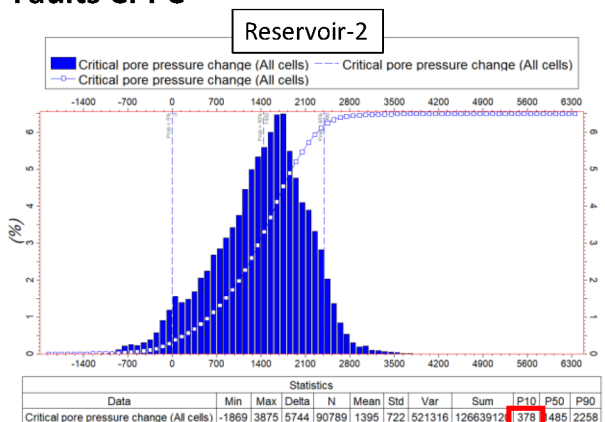


Figure 17—CPP (left) & CPPC (right) histograms in Reservoir-2 reservoir, for DFN (above) and Faults (below) in 2024.

Taking the example of Reservoir-2 in Figure 17, the maximum injection pressure (CPP) should not exceed 5,762 psi and 4619 psi to destabilize part of the DFN and Fault segments respectively. This is equivalent to a pore pressure change of 1,199 and 378 psi relative to current pore pressure. Thus, the maximum injection pressure is approximately 1,033 psi bigger than current P50 pore pressure of 4,729 psi in the case of fractures reactivation, but it is already in the limit for fault reactivation (as shown in Figure 16). Furthermore, sensitivity to stress direction confirmed that the local SHmax orientation (N60W) increases the reactivation risk compared with the regional SHmax (N40E), producing more propensity to fault reactivation and presence of critically stresses fractures (open fractures).

In general (i.e. for all the reservoirs), it can be stated that the weakest point corresponds to the fault system, with an average CPP of 5,661 psi and maximum allowed pore pressure change (CPPC) of 960 psi relative to 2024 pore pressure. These numbers are slightly bigger for the fracture network that shows an average reactivation pressure CPP of 6,261 psi and maximum allowed pore pressure change (CPPC) of 1,418 psi relative to 2024 pore pressure.

A hydrofracturing risk analysis shows an average injection pressure window of 1,706 psi (minimum stress in the cap rock minus pore pressure in the reservoir). In this case, Bottom Hole injection pressures BHP(i) should not exceed the minimum stress values. Both CPPC values for faults and fractures are smaller than the average injection pressure window, meaning that fault and fracture reactivation by shearing will occur before any hydraulic fracture by tension.

Ground level / Terrain displacement risk

Ground level subsidence also represents a geomechanics risk for wellheads, surface facilities and civil constructions (roads, buildings, etc). The analysis of surface subsidence has been detailed in works like those presented by [Dudley, 2008](#). It consists in analysing the amount of vertical displacement achieved across the field. In the area of study, the maximum amount of ground-level subsidence was -2.0 cm and uplift up to +0.74 cm, both occurring in 2055 ([Figure 18](#)). These levels of displacement do not impose failure risks to downhole completions, wellheads or surface facilities. Locations with the maximum upward and downward displacement might be selected for monitoring through GPS, fiber optics, etc.

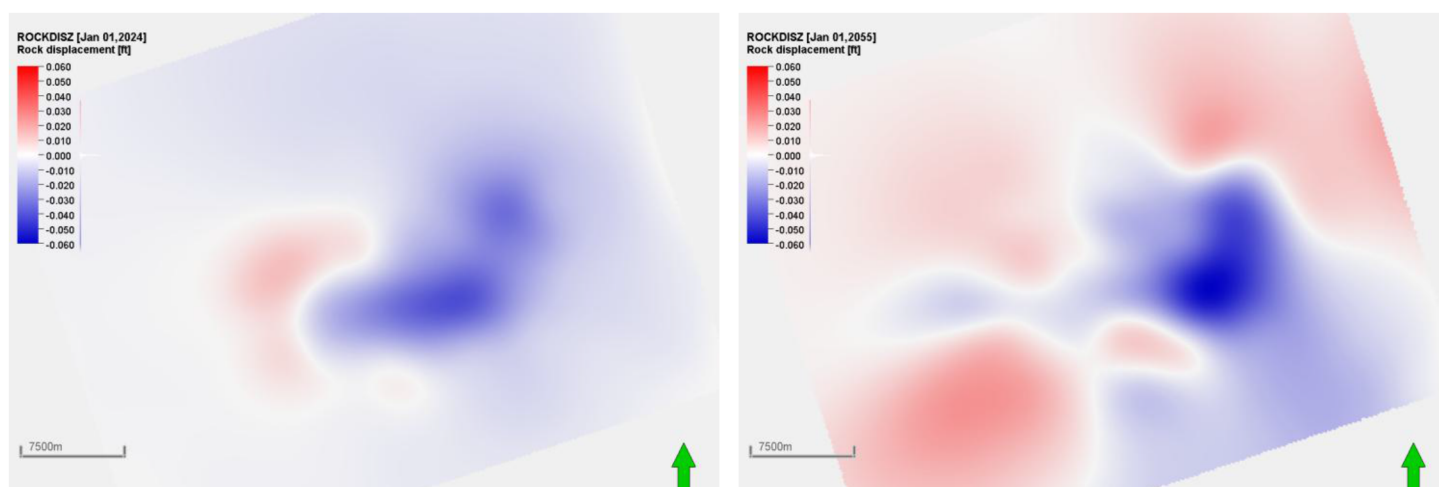


Figure 18—Ground level uplift (red) and subsidence (blue) for time step 2024 (left) and 2055 (right).

Drilling and Completion Integrity

The 4D geomechanics results are used to predict the mud window for the planned wells. An example of the mud weight window (MWW) is shown in [Figure 19](#). The MWW typically range between 10 to 12 ppg, with fracture gradients generally larger than 12 ppg. Potential breakouts are to be expected at around 10 ppg in some wells. In most wells, breakdown is below the mud weight. However, this does not represent any risk for drilling, as is mostly associated to thermal cooling rather than to hydraulic fracture creation (responsible of mud losses). Some wells are even crossing fault elements that are likely critically stressed, as represented by the discrete pink bars in track 2. Hence, these critically stresses faults might represent a hazard as they show slightly lower minimum stress compared to the surrounding intact rock. If critically stressed fractures are also encountered, as represented by the discrete pink bars in track 3, then the risk of mud losses during drilling might increase in those regions and should be keep into account during the well planning phase.

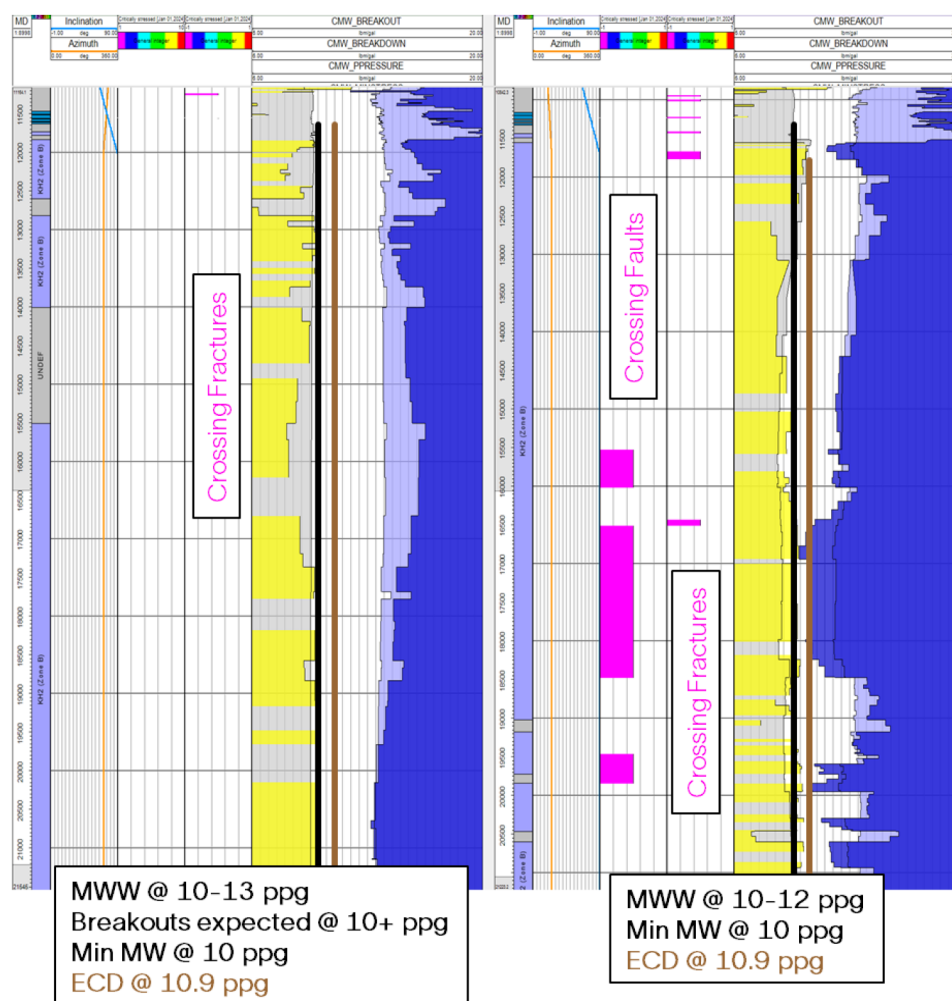


Figure 19—Wellbore stability analyses for two planned wells, with minimum mud weight (black line) and maximum equivalent circulating density (ECD in brown). Pink bars show the presence of critically stressed faults and fractures.

Since the prevailing stress regime is normal to strike slip, it can be said that wells drilled parallel to the minimum horizontal stress (S_{Hmin}) should be more stable in terms of drilling (only maybe excepting in regions with stress rotations in proximity to faults); nonetheless, it might also lead to intersect a larger number of natural fractures, requiring a close control in the amount of applied static and dynamic overbalance (MW-PP and ECD-PP), plus the use of proper bridging agents. Furthermore, the ECD should not exceed the new fracture gradient limit determined from the 4D geomechanics simulation, since as was mentioned previously, the minimum principal stress (and minimum horizontal stress) decreased in regions with heavy depletion due to poroelastic effects, impacting the mud window.

Cement and casing integrity seems not be at risk due to low reservoir displacement and ground level subsidence, high rock and cement strength, and strong casing design (some of these effects are described by Friedrich, 1998); limiting the amount of plastic deformation and potential failure to near zero. This conclusion, however, is only valid for the conditions of the proposed field development strategy.

Conclusions

Coupled 4D geomechanics modelling is a powerful tool to assess field development risks associated to caprock integrity, reservoir deformation and compaction, fracture corridor deformation and permeability evolution, fault reactivation potential and intra-reservoir communication risks, reservoir/caprock/ground level changes and wellbore integrity issues in borehole, cement and casing string.

Hydrocarbon production and WAG injection induce deformation in the rock driven by the amount of reservoir pressure increase/ decrease, the elastic properties of the rock mass and the presence of discontinuities like fractures and faults.

Fault reactivation and intra-reservoir communication is possible in regions with large DFN and Fault YIELD VALUES, especially heavily fractured zones and when injection pressure exceeds the CPPC, defined as 1,199 and 378 psi for fractures and faults respectively.

The presence of CSF might impact the intra-reservoir isolation/communication and even the borehole stability. In fact, potential fault reactivation was observed in a couple of producing wells during the history, showing correlation between the presence of CSF and the interference/communication between reservoirs. This risk can be warned by looking at faulted elements that might be critically stressed.

Pore collapse is not a risk in the study area, due to the high rock strength and relatively low effective stresses. Nonetheless, minor elastic compaction occurs, slightly affecting the porosity and permeability multipliers by -1.8% to +1.3% at the end of the FDP.

The limited changes in the reservoir pressures associated to the field development plan (FDP) produces minor deformation in the reservoir matrix and cap rock, also affected by the elastic properties. These changes limit any stress-dependent variation in the permeability of the rock; nonetheless, the contribution of the fracture system might play a significant role in the total permeability change, possibly impacting the fluid flow, injectivity and hydrocarbon production.

The fracture network is very sensitive to pressure and stress changes, producing significant alteration in the initial aperture that led to changes in the fracture permeability multipliers from -40% to +70% by 2055, potentially impacting fluid flow and reservoir connectivity.

The key geomechanics risks during the FDP in the area of study include elastic and plastic deformation of the discontinuities, destabilization of the fracture network and fault system (potential presence of CSF), ground level subsidence or heave, drilling, wellbore and completion integrity risks due the reduction in the minimum stress, and possible changes in the fluid flow pattern and intra-reservoir communication as consequence of the destabilization of fractures and faults.

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